P0457-LOOSE FUEL FILL CAP — Continued

TEST	ACTION	APPLICABILITY
6	**NOTE: A thorough visual inspection of the Evap system hoses, tubes, and connections may save time in your diagnosis. Look for any physical damage or signs of wetness at connections. The strong smell of fuel vapors may aid diagnosis also.** To continue testing, you will need Miller Tool #8404 Evaporative Emissions Leak Detector (EELD). Remove the Air supply hose from the service port or the #8404-ADP adapter. Connect the SMOKE supply tip (black hose) to the service port (if equipped) or to the #8404-ADP adapter. Set the smoke/air control switch to SMOKE. **NOTE: The flow meter indicator ball will not move at this point.** Press the remote smoke/air start button. **NOTE: Make sure that smoke has filled the EVAP system by continuing to press the remote smoke/air start button, remove the vehicle fuel cap, and wait for the smoke to exit. Once smoke is indicated reinstall the fuel cap.** **NOTE: For optimal performance, introduce smoke into the system for an additional 60 seconds; continue introducing smoke at 15 second intervals, as necessary.** While still holding the remote smoke/air start button, use the white light (#8404-CLL) to follow the EVAP system path, and look for the source of the leak indicated by exiting smoke. If a leak is concealed from view (i.e., top of fuel tank), release the remote smoke/air start button, and use the ultraviolet (UV) black light #8404-UVL and the yellow goggles 8404-20 to look for residual traces of dye that are left behind by the smoke. The exiting smoke deposits a residual fluid that is either bright green or bright yellow in color when viewed with a UV light. **NOTE: The EVAP System is divided into three zones. Possible leaking areas can be from any of the three Zones.** ZONE 1. - Fuel tank valves (control valve and inlet check valve). - Cracked, cut, warped, or missing primary fuel filler cap to fuel filler tube seal. - Out of round fuel filler tube. - Fuel filler tube to fuel filler hose connection. - Fuel filler hose to tank connection. - Fuel tank shell. - Fuel pump module attachment. - Gas cap loose or seal damaged. - Gas cap internal valve. - Damaged or disconnected EVAP system components. ZONE 2. - EVAP hoses loose or disconnected. - Damaged or disconnected EVAP system components. - Canister seals/hoses leaking. - Canister housing. - NVLD Assembly. ZONE 3. - EVAP Purge valve. - EVAP hoses loose, cracked, or disconnected. **NOTE: Carefully inspect the vent side of the EVAP Canister. Due to the filtering system in the canister the smoke or dye may or may not be visible. Introducing smoke into the filtered side of the canister may assist in locating the leak.** Was a leak found? Yes → Go To 7 No → Go To 8	All